ASR Live

Offline Real-Time Speech Recognition

离线实时语音识别 · LLM 矫正 · 多语言翻译

Installation & User Guide v3.0 · 安装与使用说明

1. System Requirements 系统要求

Item *项目*	Requirement *要求*
Hardware *硬件*	Apple Silicon Mac (M1 / M2 / M3 or later) *Apple Silicon Mac（M1 / M2 / M3 及以上）*
Memory *内存*	24 GB minimum, 48 GB recommended *最低 24GB，推荐 48GB*
macOS *系统*	13 Ventura or later *macOS 13 Ventura 或更高*
Storage *存储空间*	~11 GB for AI models *AI 模型约 11GB*
Internet *网络*	Only needed for first-time model download *仅首次下载模型需要联网*

⚠️ ASR Live runs entirely on-device. Your voice data never leaves your Mac.
⚠️ *ASR Live 完全在本地运行，语音数据不会上传到任何服务器。*

2. Installation 安装步骤

2.1 Install Homebrew 安装 Homebrew（已安装可跳过）

Open Terminal and run:
打开终端，运行以下命令：

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

2.2 Install ffmpeg 安装 ffmpeg

Required for MP3 recording:
录音保存为 MP3 格式需要此工具：

```
brew install ffmpeg
```

2.3 Set Up Python Environment 配置 Python 虚拟环境

Run these commands once in Terminal:
在终端依次运行以下命令（只需执行一次）：

```
# Create virtual environment
python3 -m venv ~/asr-env

# Activate it
source ~/asr-env/bin/activate

# Install dependencies
pip install mlx-whisper mlx-lm silero-vad sounddevice \
            fastapi "uvicorn[standard]" pywebview onnxruntime
```

> You only need to run these steps once. The virtual environment persists across reboots.
> *以上步骤只需执行一次，虚拟环境重启后依然有效。*

3. Download AI Models 下载 AI 模型

ASR Live requires two AI models: a speech recognition model (Whisper) and a language correction model (Qwen3). Download them once — they are cached locally and work offline forever after.
ASR Live 需要两个 AI 模型：语音识别模型（Whisper）和语义矫正模型（Qwen3）。下载一次后永久保存在本地，之后完全离线使用。

Speech Recognition Model — Whisper 语音识别模型

Model *模型*	Size *大小*	Speed *速度*	Accuracy *精度*	Recommended for *适用场景*
whisper-large-v3-turbo	~3 GB *约 3GB*	Fast *快*	Excellent *优秀*	Daily use ★ Recommended *日常使用 ★ 推荐*

whisper-large-v3-mlx	~6 GB *约 6GB*	Slower *稍慢*	Highest *最高*	Maximum accuracy *追求最高精度*

```
source ~/asr-env/bin/activate

# Recommended
hf download mlx-community/whisper-large-v3-turbo

# Optional — highest accuracy
hf download mlx-community/whisper-large-v3-mlx
```

Correction & Translation Model — Qwen3 矫正与翻译模型

Model *模型*	Size *大小*	Recommended for *适用场景*
Qwen3-14B-4bit	~8 GB *约 8GB*	48 GB RAM ★ Recommended *48GB 内存 ★ 推荐*
Qwen3-8B-4bit	~5 GB *约 5GB*	24 GB RAM *24GB 内存*

```
# 48 GB RAM — recommended
hf download mlx-community/Qwen3-14B-4bit

# 24 GB RAM
hf download mlx-community/Qwen3-8B-4bit
```

> If downloads are slow or failing, try the mirror: export HF_ENDPOINT=https://hf-mirror.com
> *下载缓慢或失败时，可使用镜像：export HF_ENDPOINT=https://hf-mirror.com*

4. Install ASR Live.app 安装应用

4.1 Copy to Applications 拷贝到应用程序文件夹

Open the DMG file and drag ASR Live into the Applications folder.
打开 DMG 文件，将 ASR Live 拖入 Applications（应用程序）文件夹。

4.2 Bypass Gatekeeper 绕过安全限制

Because ASR Live is not notarized by Apple, macOS may block it on first launch.
由于 ASR Live 未经 Apple 公证，首次打开可能被系统拦截。

Option A — Right-click method:
方法 A — 右键方式：

1. Right-click ASR Live in Applications → Open
2. Click Open in the dialog that appears

Option B — Terminal:
方法 B — 终端命令：

```
xattr -cr /Applications/ASR\ Live.app
```

4.3 Grant Microphone Access 授权麦克风

On first launch, macOS will request microphone permission. Click Allow.
首次启动时系统会请求麦克风权限，点击「允许」。

If no prompt appears: System Settings → Privacy & Security → Microphone → enable ASR Live.
若未弹出请求：系统设置 → 隐私与安全性 → 麦克风 → 开启 ASR Live。

5. Using ASR Live 使用说明

5.1 First Launch 首次启动

Allow 30–60 seconds on first launch for the AI models to load into memory. The status bar will show Ready when the app is fully loaded.
首次启动需要 30–60 秒加载 AI 模型，顶栏显示「就绪」后即可使用。

> Subsequent launches are faster because the models stay cached in system memory.
> *后续启动更快，因为模型已缓存在系统内存中。*

5.2 Start Transcribing 开始识别

- Click Start in the left sidebar or the top bar
  - *点击左侧「开始识别」或顶栏「● 开始」按钮*
- Speak naturally — pause briefly (≈0.6 s) at the end of each sentence
  - *正常说话，每句结束后稍停顿（约 0.6 秒）系统自动识别*
- Subtitles appear in real time: original text on top, translations below
  - *字幕实时显示：原文在上，译文在下*
- Click Stop to end the session and save the recording (if enabled)
  - *点击「停止识别」结束本次录音*

5.3 Interface Overview 界面说明

Area *区域*	Function *功能*
Top bar *顶栏*	Status, Export, Clear, Theme toggle, Settings *状态、导出、清空、主题切换、设置*
Left sidebar *左侧边栏*	Recording controls, language selection, VAD parameters, live metrics *录音控制、语言选择、VAD 参数、实时指标*
Subtitle area *字幕区域*	Real-time subtitles with waveform display *实时字幕与波形显示*
Bottom bar *底部搜索栏*	Keyword search with highlight & clear button, jump-to-latest *关键词搜索高亮、清除按钮、跳转最新*

5.4 Settings 设置说明

ASR & LLM Models ASR 与 LLM 模型

The app automatically scans locally downloaded models. Select from the dropdowns — no manual path entry required.
应用自动扫描本地已下载模型，直接从下拉菜单选择，无需手动填写路径。

Scene & Terminology 场景与术语

Enter domain-specific context and vocabulary to boost recognition accuracy for technical terms. This text is passed to both Whisper (as initial_prompt) and Qwen3 (as system context).
输入领域描述和专业词汇，显著提升专业术语识别精度。该文本同时传递给 Whisper（initial_prompt）和 Qwen3（system context）。

```
Example / 示例:
Cardiology consultation. Terms: atrial fibrillation, LVEF, CABG, PCI.
心血管科会诊。术语：心房颤动、左心室射血分数、冠状动脉旁路移植术。
```

Translation Target Languages 翻译目标语言

Select one or more output languages. When you speak a selected language, it is automatically excluded from translations to avoid duplication.
勾选一种或多种输出语言。说某语言时，该语言自动从译文中排除，避免重复显示。

Audio Input Parameters 音频输入参数

Parameter *参数*	Default *默认值*	Notes *说明*
Save Recording *录音保存*	On / 开启	Prompts native save dialog on stop *停止时弹出原生保存对话框*
MP3 Bitrate *MP3 码率*	192 kbps	Options: 64 / 128 / 192 / 320 kbps *可选：64 / 128 / 192 / 320kbps*
Silence threshold *句尾静音阈值*	0.6 s	Pause duration to trigger recognition *停顿多久后触发识别*
VAD sensitivity *VAD 灵敏度*	0.45	Higher = less sensitive; noisy env: 0.6–0.7 *越高越不敏感；嘈杂环境建议 0.6–0.7*
Max sentence length *最长单句时长*	20 s	Force-cuts after this duration *超时强制切断*
Microphone *麦克风设备*	System default *系统默认*	Click ↻ to rescan devices *点 ↻ 刷新已连接设备*

5.5 Export 导出

Click Export ▾ in the top bar. Choose a language filter and format, then a native save dialog appears.
点击顶栏「导出 ▾」，选择语言筛选和格式，弹出原生保存面板。

Format *格式*	Description *说明*
TXT *纯文本*	Plain text with timestamps *每条含时间戳的纯文本*
SRT *字幕文件*	Standard subtitle format for video editors *标准字幕格式，可导入视频软件*
JSON *结构化数据*	Full metadata: timestamps, latency, all translations *含时间戳、时滞、原文和所有译文*
Markdown *Markdown*	Paste into Obsidian, Notion, etc. *适合粘贴到 Obsidian、Notion 等笔记工具*

Language filter: All (mixed), Original text (corrected), Original text (raw ASR), or any single language. When a single language is selected, both original-language entries and translated entries are included.
语言筛选：全部混合、仅原文（矫正后）、仅原文（ASR 原始），或任意单一语言。选择单一语言时，作为原文或译文的条目均会包含。

6. Troubleshooting 常见问题

Issue *问题*	Solution *解决方法*
Window shows "Connecting" for a long time *窗口长时间显示「连接中」*	Wait 30–60 s for models to load. The status bar shows Ready when done. *等待 30–60 秒，顶栏显示「就绪」后即可操作。*
No translations appear *没有翻译输出*	Open Settings, check target languages, click Save Settings. *打开设置，勾选翻译目标语言并保存。若无效，删除 ~/.asrlive_settings.json 后重启。*
Repeated words (e.g. "nope nope nope…") *出现大量重复词*	Whisper hallucination — built-in filter handles most cases. Try raising VAD sensitivity to 0.6–0.7. *Whisper 幻觉，程序已内置过滤。可调高 VAD 灵敏度至 0.6–0.7。*
LLM correction is slow (>2 s) *LLM 矫正时间过长*	Switch to Qwen3-8B-4bit in Settings, or reduce the number of target languages.

	在设置中换用 Qwen3-8B-4bit，或减少翻译目标语言数量。
Recording not saved *录音没有保存*	Ensure ffmpeg is installed (brew install ffmpeg) and recording is enabled in Settings. *确认已安装 ffmpeg（brew install ffmpeg），且设置中录音开关已开启。*
Microphone not working *麦克风无法使用*	System Settings → Privacy & Security → Microphone → enable ASR Live. *系统设置 → 隐私与安全性 → 麦克风 → 开启 ASR Live。*
No module named 'onnxruntime' *缺少 onnxruntime 模块*	Run: pip install onnxruntime *运行：pip install onnxruntime*